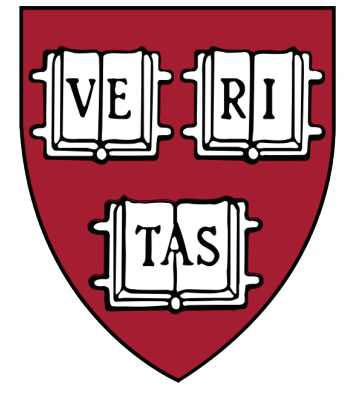




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Measuring and Controlling Solution Degeneracy across Task-trained Recurrent Neural Networks

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NEURAL INFORMATION
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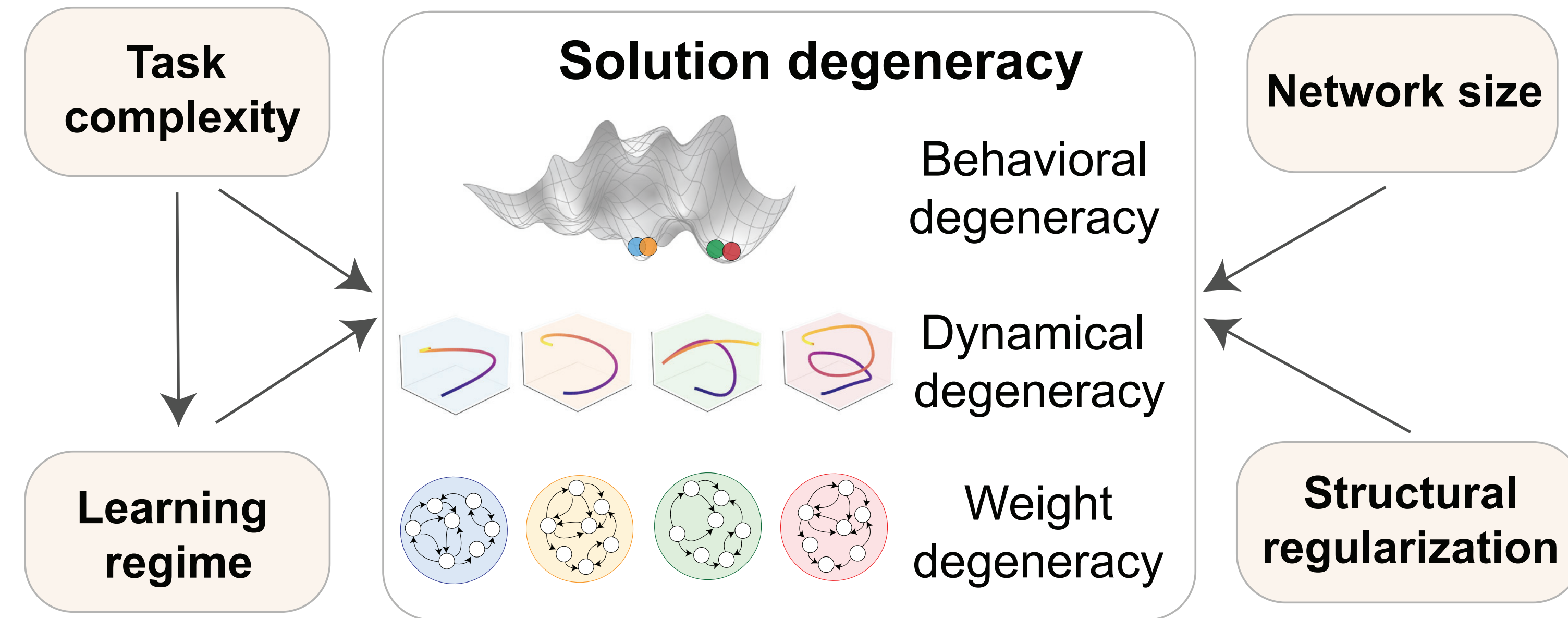
¹ Harvard University

² Kempner Institute

³ EPFL

Networks performing equally well can
show strikingly different internal solutions [1-7].

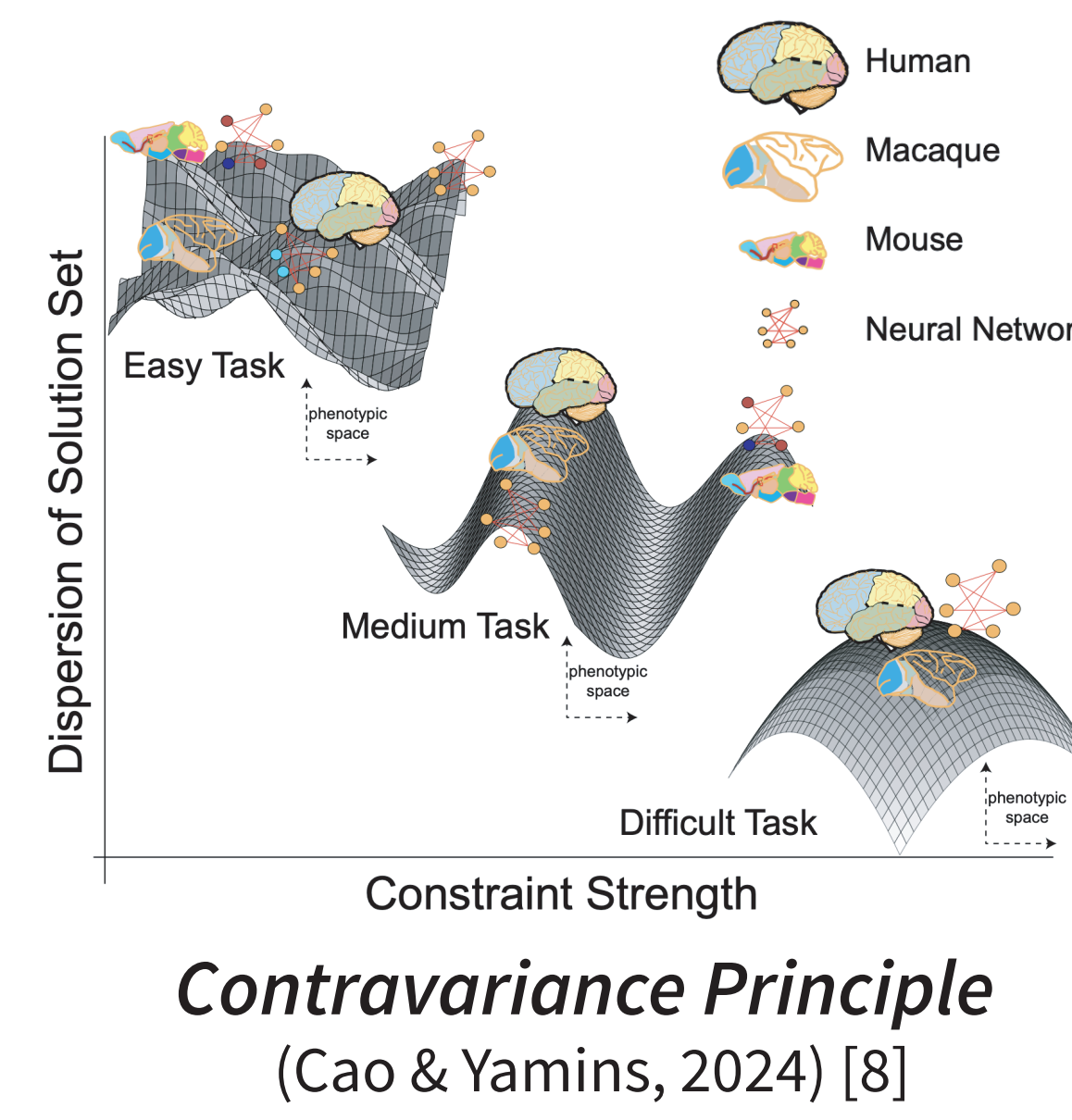
**What factors govern the solution degeneracy
across independently trained RNNs?**



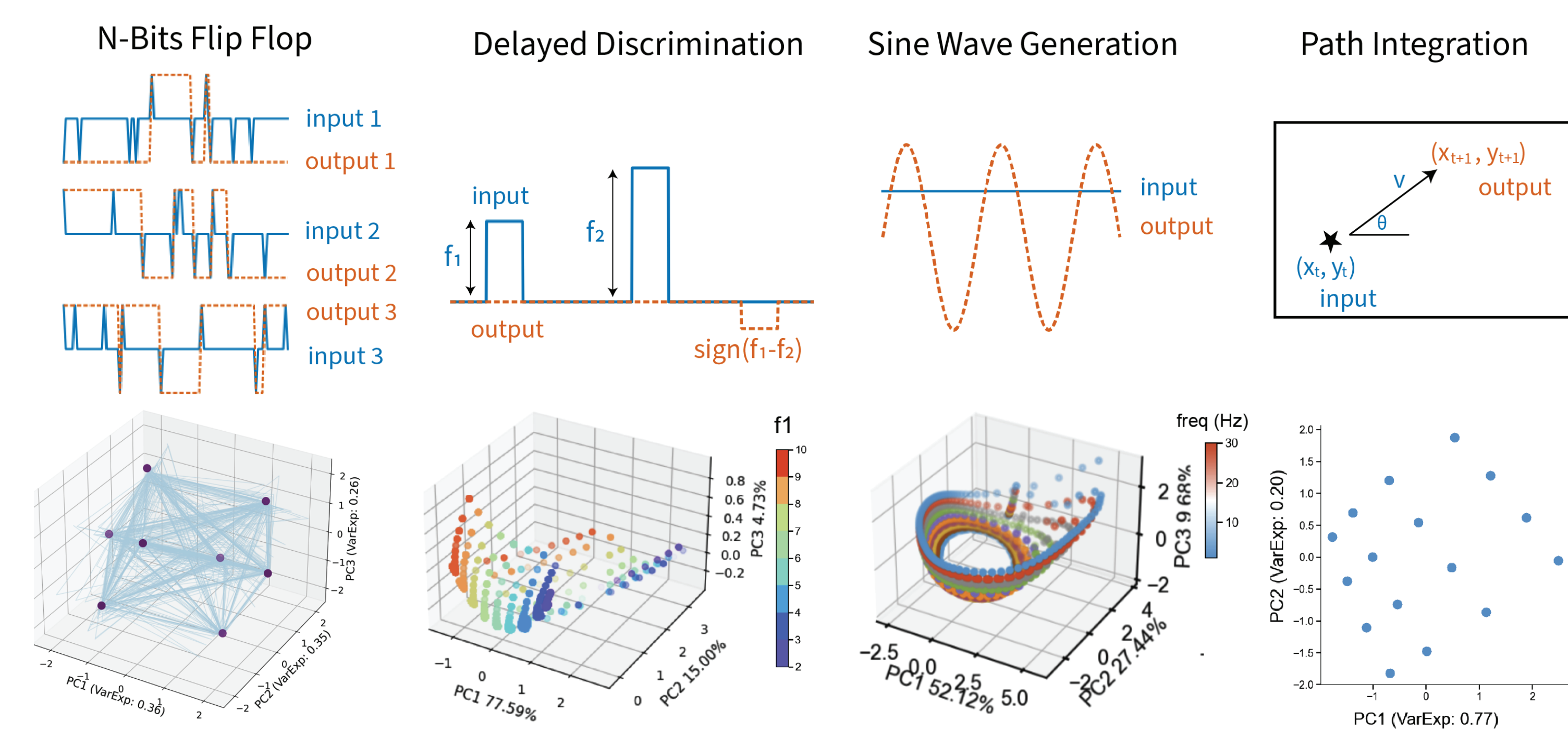
Highlights

Factor	Dynamics	Weights	Behavior
Higher Task complexity	↓	↑	↓
More Feature learning	↓	↑	↓
Larger Network size	↓	↓	↓
Regularization	↓	↓	↓

- A unified framework for quantifying solution degeneracy at behavior, neural dynamics, and weight level
- Four factors that shape the solution landscape, providing practical knobs for tuning the variability across RNN solutions
- Empirically validate the **Contravariance Principle**
- Identify when degeneracy at the dynamical and weight level goes together, or in opposite directions



Task suite



A unified framework to quantify solution degeneracy

Behavioral degeneracy: Variability of the Out-Of-Distribution (OOD) performance, quantified by the standard deviation of MSE loss on the temporal generalization condition

Dynamical degeneracy: Distance between neural trajectories, quantified by Dynamical Similarity Analysis [9].

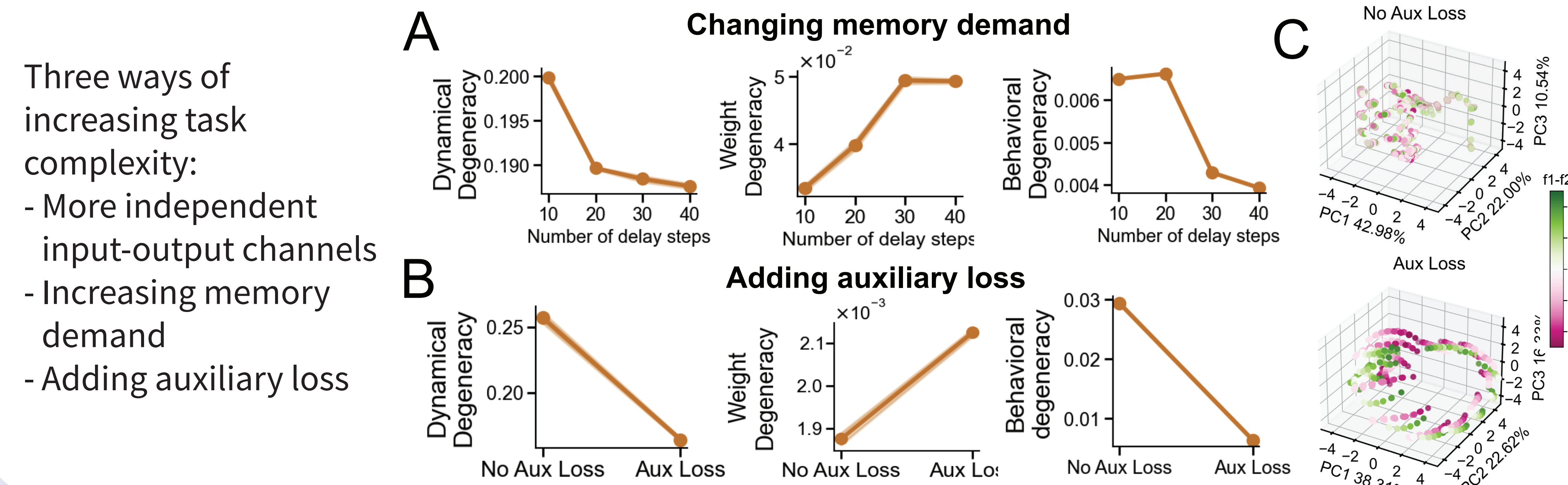
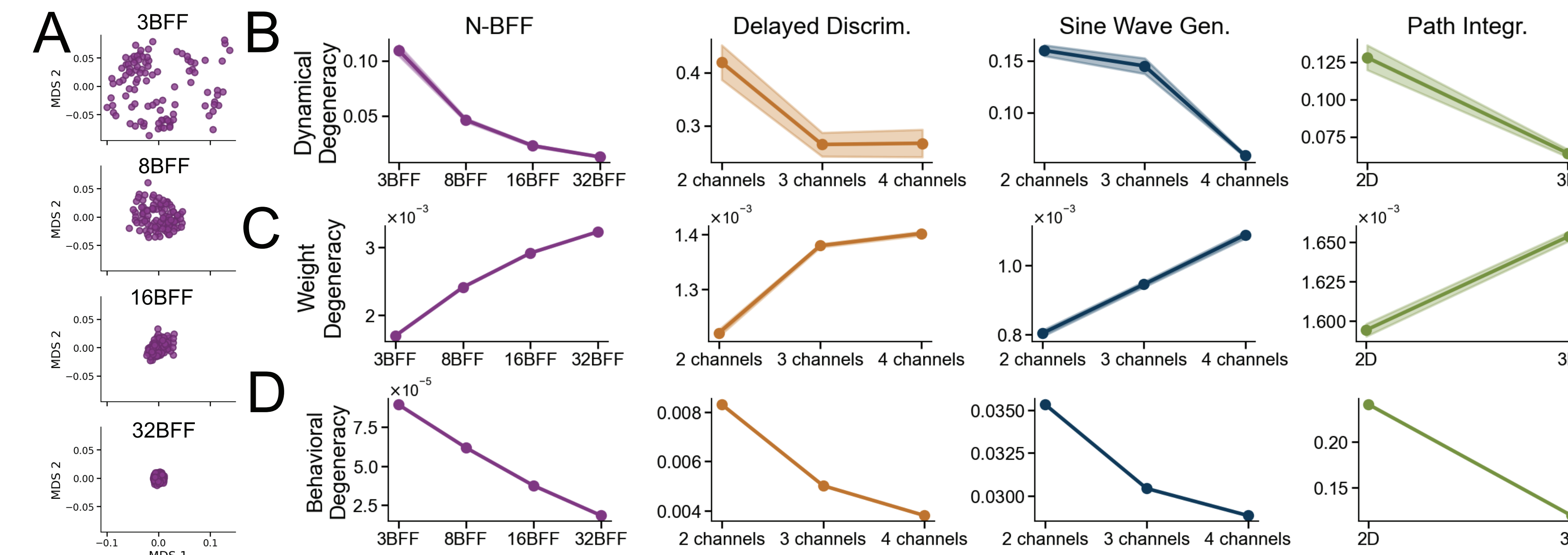
$$\tilde{x}_{t+1} = A_x \tilde{x}_t \quad \tilde{y}_{t+1} = A_y \tilde{y}_t \quad DSA(A_x, A_y) = \min_{C \in O(n)} \|A_x - CA_y C^{-1}\|_F$$

Weight degeneracy: Permutation-invariant Frobenius distance between recurrent weights [10]

$$d_{PIF}(W_1, W_2) = \min_{P \in \mathcal{P}(n)} \|W_1 - P^T W_2 P\|_F$$

Task complexity

When trained on harder tasks, RNNs converge to more similar neural dynamics and OOD behavior, but their weight configurations diverge.

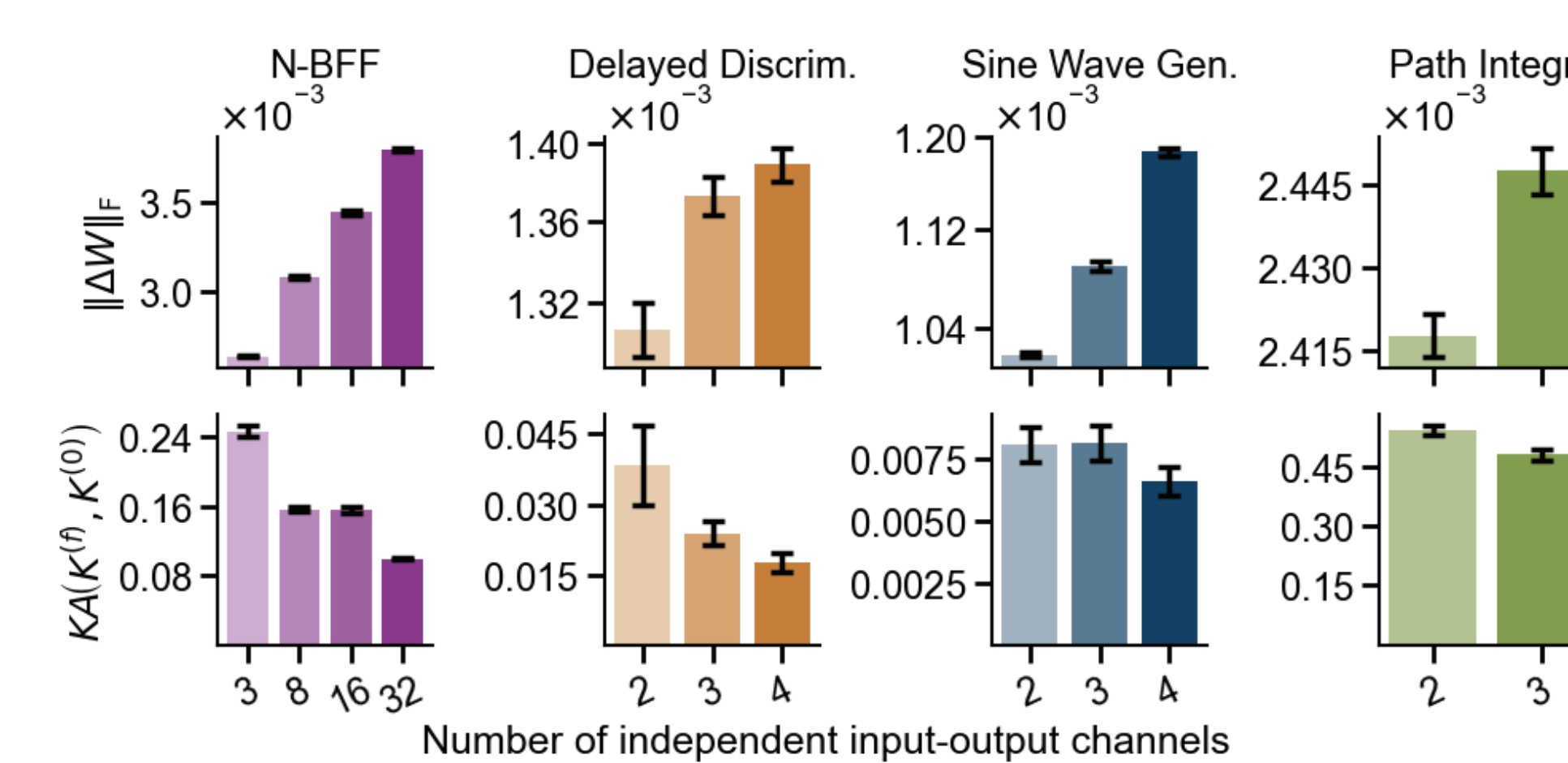


Harder tasks induce more feature learning

An increased weight dispersion for harder tasks reflect **stronger feature learning** in the networks [11]

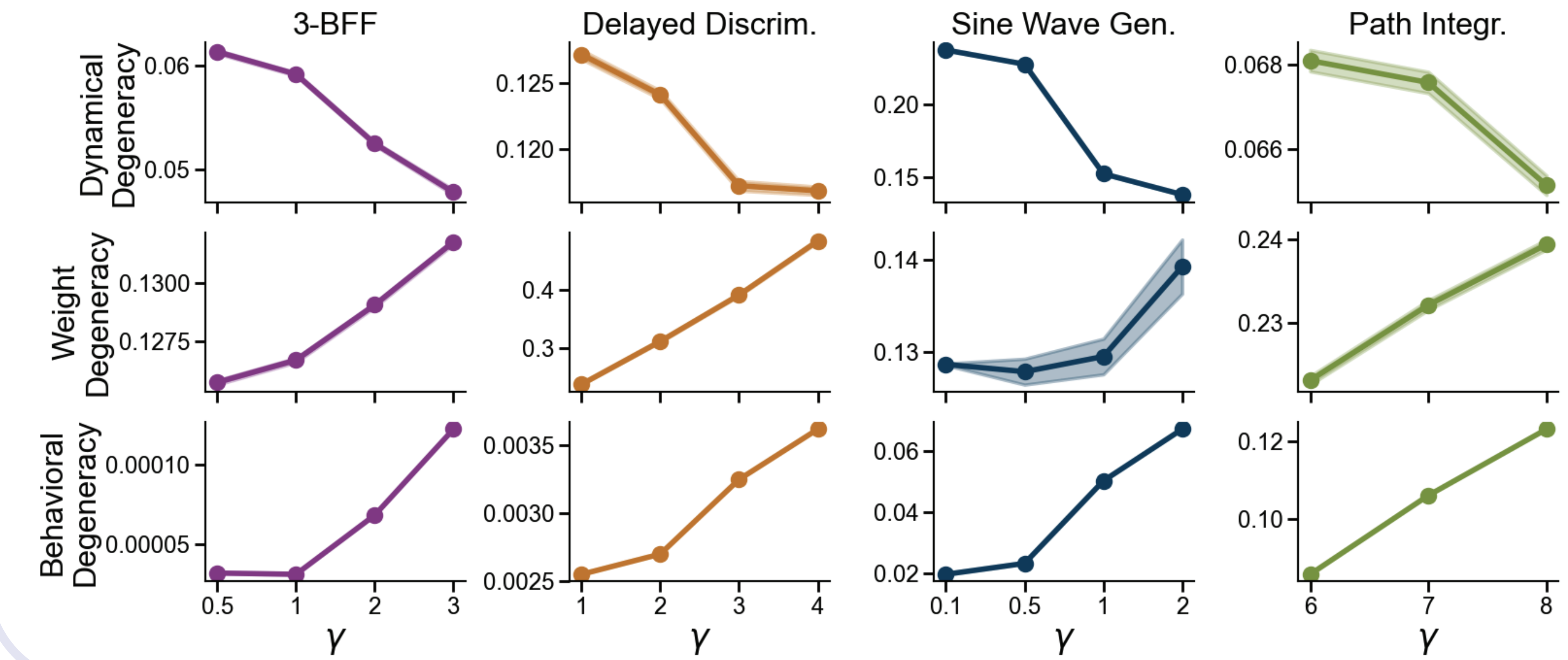
Use *Maximal Update Parameterization* to directly control for feature learning strength [12], where larger gamma -> stronger feature learning.

$$f(t) = \frac{1}{N\gamma} W_{\text{readout}} \phi(h(t))$$



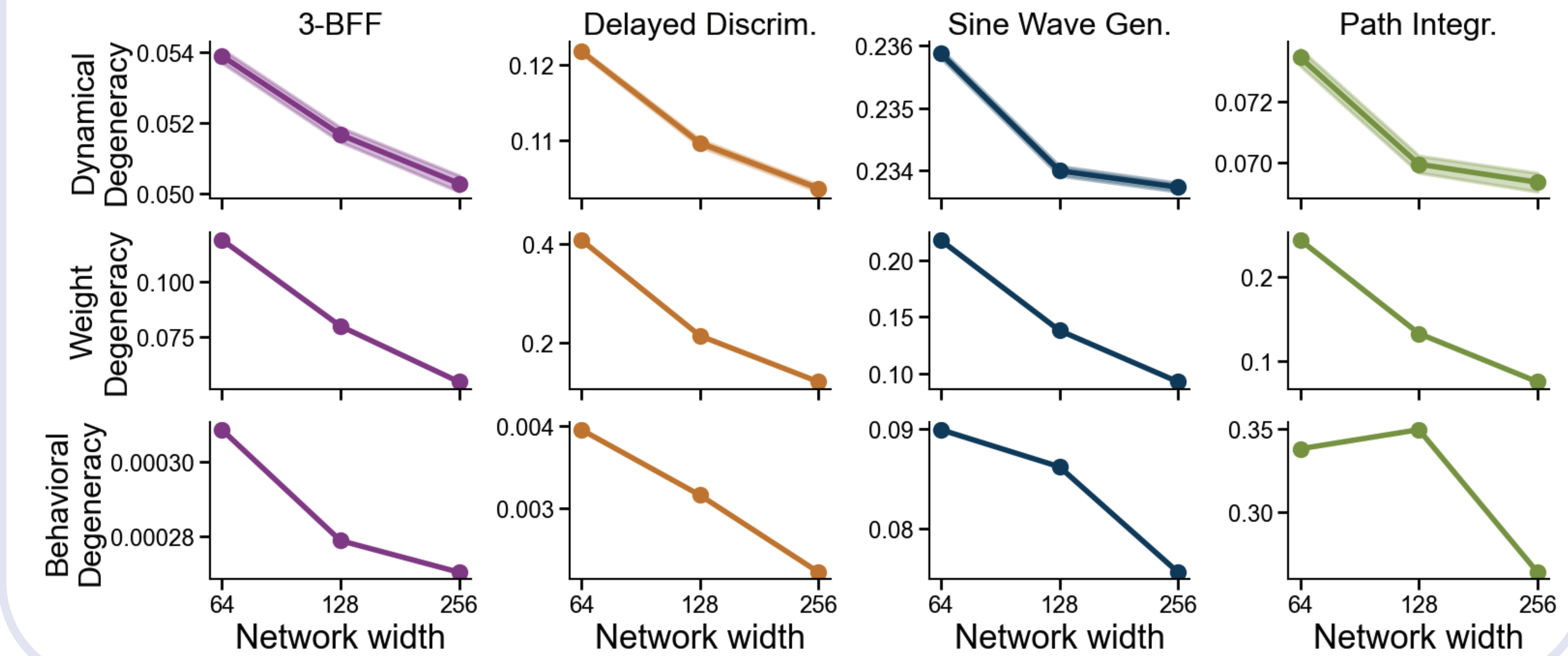
Feature learning

Stronger feature learning induces more consistent neural dynamics, but lead to more divergent weight and OOD behavioral responses.



Network size

Larger networks yield more consistent solutions across levels.



Regularization

Low-rank and sparsity regularization reduce degeneracy across levels.

Low-rank: $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{rank}} \sum_{i=1}^r \sigma_i$

$W_{\text{rec}} = U \Sigma V^T$ where $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$.

Sparsity (L1): $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\ell_1} \sum_i |w_i|$

